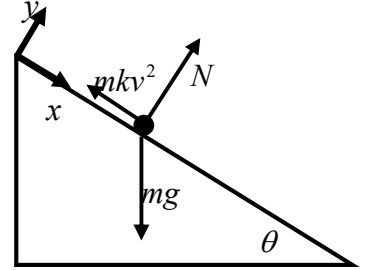


University Physics Quiz 2

Name: _____ SID: _____ School/Department: _____

1. A particle of mass m slides down an inclined plane, subject to a constant vertically downward force of magnitude mg . The motion is resisted by a force mkv^2 , k being a positive constant and v the particle's speed. Show that the time required to move a distance d after starting from rest is

$$t = \frac{\cosh^{-1}(e^{kd})}{(kg \sin \theta)^{\frac{1}{2}}}$$



where θ is the angle of inclination of the plane.

Solution: By introducing the coordinates as shown in the figure, the equation of motion can be written as

$$m\ddot{\mathbf{r}} = mg \sin \theta \mathbf{i} - mg \cos \theta \mathbf{j} - mkv^2 \mathbf{i} + N \mathbf{j}$$

Since the particle moves along the plane, we have

$$N - mg \cos \theta = 0.$$

The equation of motion can be rewritten as

$$\ddot{\mathbf{r}} = (g \sin \theta - kv^2) \mathbf{i}.$$

Let $\mathbf{r} = x\mathbf{i}$, where x is the distance the particle travels down the inclined plane. Thus

$$\frac{d^2x}{dt^2} = g \sin \theta - kv^2 \quad \text{or} \quad \frac{dv}{dt} = g \sin \theta - kv^2.$$

Separating the variables of above equation gives

$$\frac{dv}{\frac{g \sin \theta}{k} - v^2} = k dt \quad \text{or} \quad \frac{dv}{\lambda^2 - v^2} = k dt, \quad \text{where } \lambda = \sqrt{\frac{g \sin \theta}{k}}.$$

Integrating both sides

$$\int_0^v \frac{dv}{\lambda^2 - v^2} = \int_0^t k dt.$$

we obtain

$$\frac{1}{2\lambda} \ln \frac{\lambda + v}{\lambda - v} = kt \quad \text{or} \quad \frac{\lambda + v}{\lambda - v} = e^{2\lambda kt} \quad \text{or} \quad v = \lambda \frac{e^{2\lambda kt} - 1}{e^{2\lambda kt} + 1} = \lambda \tanh(\lambda kt).$$

Integrating above equation

$$x = \int_0^x dx = \int_0^t \lambda \tanh(\lambda kt) dt = \frac{1}{k} \int_0^t \tanh(\lambda kt) d(\lambda kt) = \frac{1}{k} \ln[\cosh(\lambda kt)].$$

When $x = d$,

$$kd = \ln[\cosh(\lambda kt)], \quad e^{kd} = \cosh(\lambda kt), \quad t = \frac{\cosh^{-1}(e^{kd})}{\lambda k} = \frac{\cosh^{-1}(e^{kd})}{(kg \sin \theta)^{\frac{1}{2}}}.$$

((Hyperbolic functions:

Hyperbolic sine: $\sinh(x) = \frac{e^x - e^{-x}}{2}.$

Hyperbolic cosine: $\cosh(x) = \frac{e^x + e^{-x}}{2}.$

Hyperbolic tangent: $\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{e^{2x} - 1}{e^{2x} + 1}.$

$$\int \tanh(x) dx = \int \frac{e^x - e^{-x}}{e^x + e^{-x}} dx = \int \frac{d(e^x + e^{-x})}{e^x + e^{-x}} = \int \frac{d(\cosh(x))}{\cosh(x)} = \ln(\cosh(x)) + C.$$

2. A spherical raindrop leaves a cloud with initial radius r_0 and negligible speed. As it passes through the atmosphere, its mass is increased at a rate proportional to the product of its surface area and speed v . Assume the density of the raindrop is constant and no forces act on the raindrop except gravity. The distance below the cloud is denoted as x .

(a) Show that the radius r of the raindrop increases linearly with x .

(b) If r_0 is very small, show that the acceleration of the raindrop is approximately $g - 3v^2/x$.

(c) Show also, when r_0 is very small, that the equation of motion of the raindrop can be written in the form $d(v^2)/dx + 6v^2/x = 2g$.

(d) Verify that the equation in (c) has solution $v^2 = 2gx/7$, and show that the acceleration of the raindrop is $g/7$.

Solution: (a) According to the problem

$$\frac{dm}{dt} = k(4\pi r^2)v, \quad m = \frac{4}{3}\pi r^3 \rho.$$

Thus

$$\frac{dm}{dt} = \frac{d\left(\frac{4}{3}\pi r^3 \rho\right)}{dr} = \rho 4\pi r^2 \frac{dr}{dt} = k(4\pi r^2)v \quad \text{or} \quad \frac{dr}{dt} = \frac{k}{\rho} v = \frac{k}{\rho} \frac{dx}{dt} \quad \text{or} \quad dr = \frac{k}{\rho} dx.$$

Integration both sides of above equation

$$\int_{r_0}^r dr = \frac{k}{\rho} \int_0^x dx, \quad r = r_0 + \frac{k}{\rho} x. \quad (1)$$

(b) The equation of motion of the rain drop is

$$\frac{d(mv)}{dt} = m \frac{dv}{dt} + v \frac{dm}{dt} = mg.$$

Using $\frac{dm}{dt} = k(4\pi r^2)v$, we have

$$m \frac{dv}{dt} = mg - k(4\pi r^2)v^2.$$

Thus

$$\frac{dv}{dt} = g - \frac{k(4\pi r^2)}{\frac{4}{3}\pi r^3 \rho} v^2 = g - \frac{3k}{\rho} \frac{v^2}{r}.$$

If r_0 is very small, $r \approx \frac{k}{\rho} x$. Substituting it into above equation gives

$$\frac{dv}{dt} = g - \frac{3v^2}{x}. \quad (2)$$

(c) Noting that

$$\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx} = \frac{1}{2} \frac{d(v^2)}{dx},$$

the above equation can be rewritten as

$$\frac{d(v^2)}{dx} + \frac{6v^2}{x} = 2g \quad (3)$$

(d) Substituting $v^2 = \frac{2gx}{7}$ into above equation, we have:

$$\text{Left side} = \frac{d\left(\frac{2gx}{7}\right)}{dx} + \frac{6\left(\frac{2gx}{7}\right)}{x} = \frac{2g}{7} + \frac{12g}{7} = 2g = \text{Right side.}$$

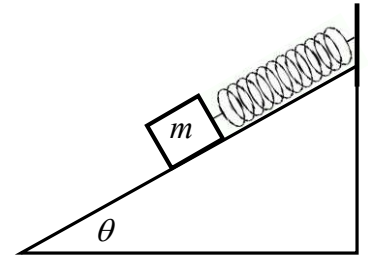
Substituting $v^2 = 2gx/7$ into (2), we have

$$dv/dt = g - \frac{3(2gx/7)}{x} = \frac{g}{7}.$$

3. A block of mass m rests on an incline that makes an angle of θ with the horizontal plane. The static and kinetic friction coefficients between the block and the surface of incline are μ_s and μ_k respectively ($\mu_s > \mu_k$). The block is attached to a massless spring of spring constant k and relaxed length l , as shown in the figure.

The block is pulled so that spring is extended to a length $l + x_{\max}$, where $x_{\max}$ is the maximum extension of spring with which the block remains stationary when released.

The block is released and then given a tiny kick and it starts moving up.



(a) Find the maximum extension $x_{\max}$.

(b) For what value of the extension of the spring x , the block reaches its maximum speed?

(c) As the block moves up-hill, the spring gets shorter. Show that the condition for the spring can become shorter than its relaxed length l is $\mu_s > 2\mu_k + \tan \theta$.

Solution: (a) The free-body diagram: (The position of the block where the spring is relaxed is taken as the original point.)

The equilibrium of requires

$$x: -kx + F_{sf} + mg \sin \theta = 0,$$

$$y: F_N - mg \cos \theta = 0.$$

$$F_{sf} \leq \mu_s F_N$$

From the above equations, we obtain

$$kx \leq \mu_s mg \cos \theta + mg \sin \theta \quad \text{or} \quad x \leq \frac{mg}{k} (\mu_s \cos \theta + \sin \theta).$$

$$\text{Therefore, } x_{\max} = \frac{mg}{k} (\mu_s \cos \theta + \sin \theta)$$

(b) The equation of motion is

$$-kx + \mu_k mg \cos \theta + mg \sin \theta = ma.$$

At the point the acceleration $a = 0$, the block reaches the maximum speed. Thus, the extension for the maximum speed is

$$x = \frac{mg}{k} (\mu_k \cos \theta + \sin \theta).$$

(c) (Solution 1) The equation of motion is

$$m \frac{dv}{dt} = -kx + \mu_k mg \cos \theta + mg \sin \theta \quad \text{or} \quad m \frac{dv}{dx} \cdot \frac{dx}{dt} = -kx + \mu_k mg \cos \theta + mg \sin \theta$$

$$\text{or} \quad mv dv = (-kx + \mu_k mg \cos \theta + mg \sin \theta) dx.$$

Integrating both sides of the equation from the initial point $x = x_{\max}$ to the highest point the block can reach

$x = x_h$ (where the speed of the block is zero), we have

$$\frac{1}{2}mv^2 \Big|_0^0 = \left[-\frac{1}{2}kx^2 + mg(\mu_k \cos \theta + \sin \theta)x \right]_{x=x_{\max}}^{x=x_h}$$

or

$$-\frac{1}{2}k(x_h^2 - x_{\max}^2) + mg(\mu_k \cos \theta + \sin \theta)(x_h - x_{\max}) = 0$$

or

$$-\frac{1}{2}k(x_h + x_{\max}) + mg(\mu_k \cos \theta + \sin \theta) = 0.$$

The condition for the spring can become shorter than its relaxed length l is $x_h < 0$. Thus we have

$$-\frac{1}{2}kx_{\max} + mg(\mu_k \cos \theta + \sin \theta) < 0 \quad \text{or} \quad x_{\max} > 2\frac{mg}{k}(\mu_k \cos \theta + \sin \theta).$$

By using the result of (a), we have

$$\frac{mg}{k}(\mu_s \cos \theta + \sin \theta) > 2\frac{mg}{k}(\mu_k \cos \theta + \sin \theta).$$

Thus $\mu_s > 2\mu_k + \tan \theta$.

(Solution2) Since the kinetic energies of the block at the initial point $x = x_{\max}$ and the highest point $x = x_h$ are both zero, then in the process the block moves from $x_{\max}$ to x_h , the decrease of the elastic potential energy of the spring is equivalent to the increase of the gravitational potential energy plus the work done by the frictional force. Therefore, we obtain

$$\frac{1}{2}kx_{\max}^2 - \frac{1}{2}kx_h^2 = mg(x_{\max} - x_h)\sin \theta + mg\mu_k \cos \theta(x_{\max} - x_h)$$

or

$$\frac{1}{2}k(x_{\max} + x_h) = mg(\mu_k \cos \theta + \sin \theta).$$

Since $x_h < 0$, we have $\frac{1}{2}kx_{\max} > mg(\mu_k \cos \theta + \sin \theta)$. Using the result of (a), we obtain $\mu_s > 2\mu_k + \tan \theta$.

4. A bead of mass M moves without friction along a stationary rod that lies along the horizontal x-axis. A pendulum of length L is suspended from this bead with a mass m at its end. The pendulum swings in the xy-plane, where y is the vertical axis. Assuming small oscillations, what is the frequency at which the pendulum oscillates? (Hint: To write the equation of motion of the pendulum in the noninertial frame of the bead.)

Solution: The equation of motion of the pendulum is

$$e_\rho: \quad mg \cos \theta - F_T - m\ddot{x} \sin \theta = m(-L\dot{\theta}^2)$$

(1)

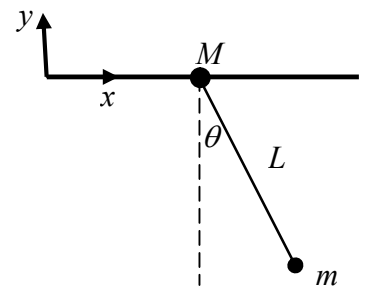
$$e_\theta: \quad -mg \sin \theta - m\ddot{x} \cos \theta = mL\ddot{\theta} \quad (2)$$

The equation of motion of the bead is

$$M\ddot{x} = F_T \sin \theta \quad (3)$$

(3) - (1) $\times \sin \theta$ - (1) $\times \cos \theta$:

$$(M + m)\ddot{x} = mL\dot{\theta}^2 \sin \theta - mL\ddot{\theta} \cos \theta \quad (4)$$



For small oscillations, using the approximations $\sin \theta \approx \theta$, $\cos \theta \approx 1$ and neglecting the high order small quantity, equations (2) and (4) can be approximated as

$$-mg\theta - m\ddot{x} = mL\ddot{\theta} \quad (5)$$

and

$$(M + m)\ddot{x} = -mL\ddot{\theta} \quad (6)$$

Substituting $\ddot{x} = -\frac{m}{M + m}L\ddot{\theta}$ into (5) and after some algebra, we have the small oscillation equation

$$\ddot{\theta} + \frac{M + m}{M} \frac{g}{L} \theta = 0$$

Thus $\omega = \sqrt{\frac{M + m}{M}} \sqrt{\frac{g}{L}}$.